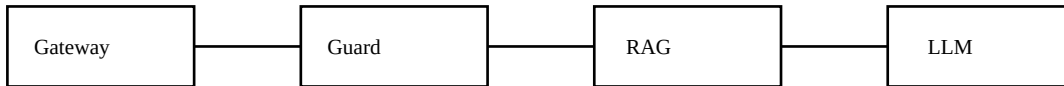


AI Safety Gateway

Software Architecture Specification (SAS) • Product Blueprint v1.0

This document is intentionally written as an implementation-ready architecture specification that Cursor AI or other coding agents can use as the single source of truth before generating the repository.

Page 2: Executive Summary



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Executive Summary**. Cursor AI should treat this chapter as authoritative when generating code.

- Build a centralized AI Gateway that protects every LLM request.
- Support RAG, Guardrails, Evaluations and Observability.
- Docker-first development: everything runs through Docker Compose.
- Enterprise-ready architecture with future Kubernetes deployment.

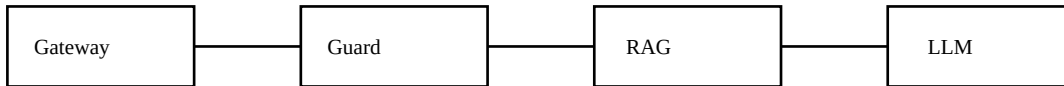
*****Acceptance Criteria*****

Every implementation generated from this chapter must remain Docker-compatible, expose health checks, structured logging, environment-based configuration and production-safe defaults.

*****AI Agent Instructions*****

Generate interfaces before implementations, keep configuration externalized, include README documentation and preserve service boundaries.

Page 3: Problem Statement



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Problem Statement**. Cursor AI should treat this chapter as authoritative when generating code.

- AI teams repeatedly rebuild authentication, prompt protection, logging and retrieval.
- Applications expose inconsistent security.
- RAG quality is difficult to monitor.
- No centralized governance exists.

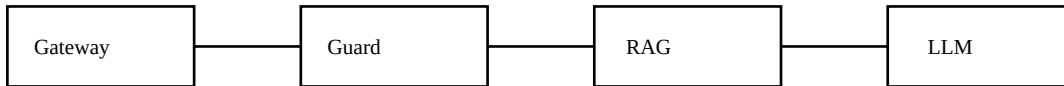
Acceptance Criteria

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AI Agent Instructions

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Page 4: Product Vision



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Product Vision**. Cursor AI should treat this chapter as authoritative when generating code.

- One gateway serving HR, Engineering, Legal and Support assistants.
- Provider-agnostic model routing.
- Centralized policy enforcement.

Acceptance Criteria

Every implementation generated from this chapter must remain Docker-compatible, expose health checks, structured logging, environment-based configuration and production-safe defaults.

AI Agent Instructions

Generate interfaces before implementations, keep configuration externalized, include README documentation and preserve service boundaries.

Page 5: Success Metrics



This chapter defines the architectural decisions, responsibilities and implementation boundaries for ****Success Metrics****. Cursor AI should treat this chapter as authoritative when generating code.

- P95 latency below 2 seconds.
- Faithfulness above 95%.
- Zero unauthorized document exposure.
- Full audit trail.

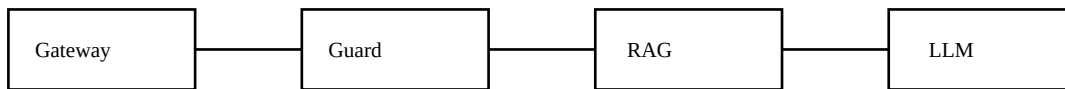
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****AI Agent Instructions****

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Page 6: Personas



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Personas**. Cursor AI should treat this chapter as authoritative when generating code.

- Platform Engineers
- AI Engineers
- Backend Engineers
- Security Teams
- Product Teams

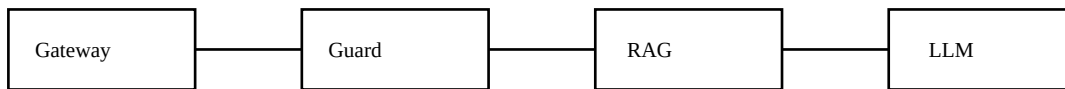
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Page 7: Core Architecture



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Core Architecture**. Cursor AI should treat this chapter as authoritative when generating code.

- Client → Gateway → Guardrails → Retriever → LLM → Verification → Response
- Every request receives tracing and evaluation.

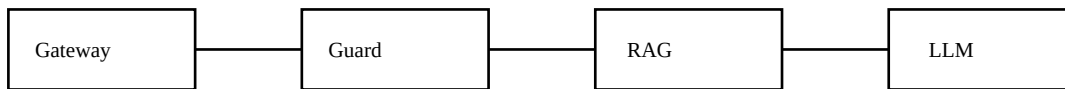
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*****AI Agent Instructions*****

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Page 8: Tech Stack



This chapter defines the architectural decisions, responsibilities and implementation boundaries for ****Tech Stack****. Cursor AI should treat this chapter as authoritative when generating code.

- FastAPI
- LangGraph
- PostgreSQL
- Redis
- Kafka
- OpenTelemetry
- Prometheus
- Grafana
- Docker Compose
- Kubernetes (future)

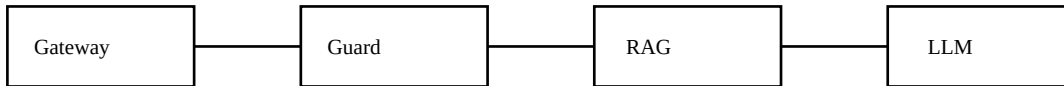
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Page 9: Docker Philosophy



This chapter defines the architectural decisions, responsibilities and implementation boundaries for ****Docker Philosophy****. Cursor AI should treat this chapter as authoritative when generating code.

- No local PostgreSQL installation.
- No local Redis installation.
- No local Kafka installation.
- Single docker compose command.

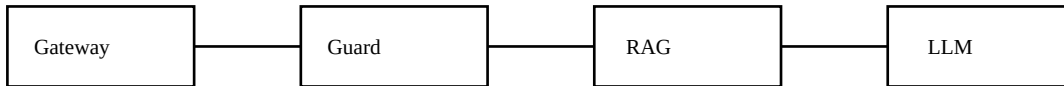
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****AI Agent Instructions****

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Page 10: Repository Structure



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Repository Structure**. Cursor AI should treat this chapter as authoritative when generating code.

- apps/gateway
- apps/worker
- services/rag
- services/guardrails
- services/evals
- infrastructure/docker
- docs
- tests

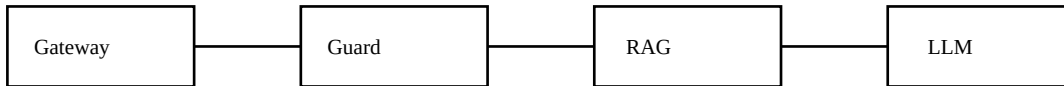
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Page 11: Gateway Service



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Gateway Service**. Cursor AI should treat this chapter as authoritative when generating code.

- JWT Authentication
- API Keys
- Streaming
- Rate Limiting
- Request Validation

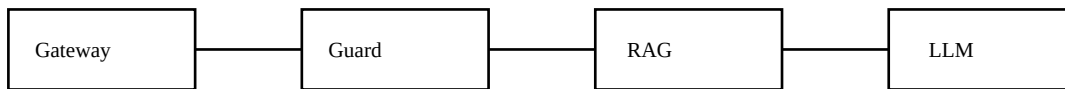
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*****AI Agent Instructions*****

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Page 12: Authentication & Authorization



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Authentication & Authorization**. Cursor AI should treat this chapter as authoritative when generating code.

- JWT verification
- RBAC
- Tenant Isolation
- API Keys
- Audit Logs

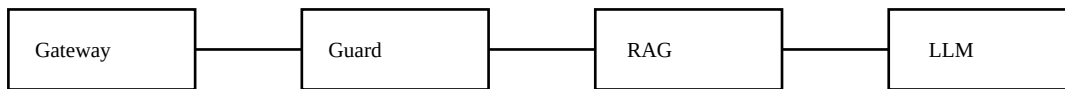
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AI Agent Instructions

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Page 13: Input Guardrails



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Input Guardrails**. Cursor AI should treat this chapter as authoritative when generating code.

- Prompt Injection Detection
- Jailbreak Detection
- Token Limits
- Content Moderation

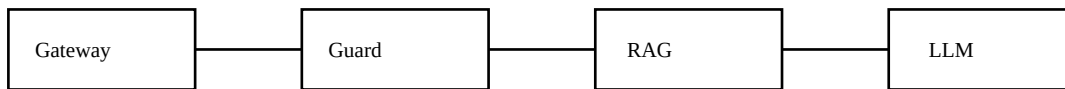
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AI Agent Instructions

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Page 14: Retrieval Guardrails



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Retrieval Guardrails**. Cursor AI should treat this chapter as authoritative when generating code.

- RBAC Filtering
- Secret Removal
- Tenant Filtering
- Sensitive Chunk Filtering

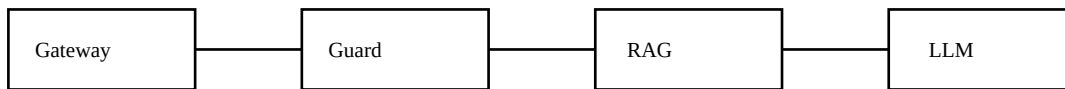
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AI Agent Instructions

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Page 15: Output Guardrails



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Output Guardrails**. Cursor AI should treat this chapter as authoritative when generating code.

- Hallucination Detection
- Citation Verification
- PII Redaction
- Confidence Scoring

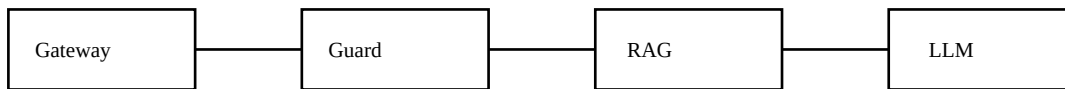
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AI Agent Instructions

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Page 16: RAG Pipeline



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **RAG Pipeline**. Cursor AI should treat this chapter as authoritative when generating code.

- Query Classification
- Hybrid Search
- BM25 + Vector
- Reranking
- Context Filtering
- Grounded Generation

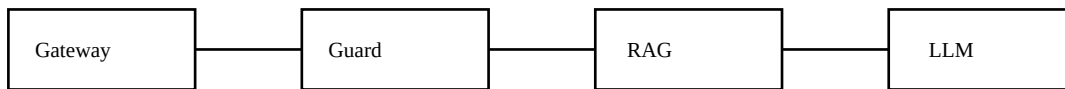
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*****AI Agent Instructions*****

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Page 17: Evaluation Framework



This chapter defines the architectural decisions, responsibilities and implementation boundaries for ****Evaluation Framework****. Cursor AI should treat this chapter as authoritative when generating code.

- Golden Dataset
- Faithfulness
- Context Recall
- Context Precision
- Answer Correctness
- Latency Tracking

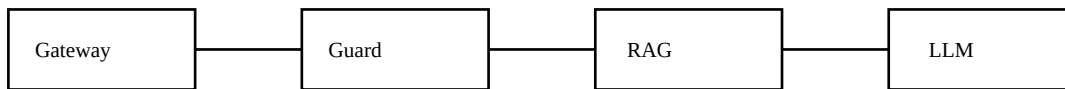
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Page 18: Self-Healing Pipeline



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Self-Healing Pipeline**. Cursor AI should treat this chapter as authoritative when generating code.

- Retry with more context
- Rewrite query
- Change reranker
- Escalate failures

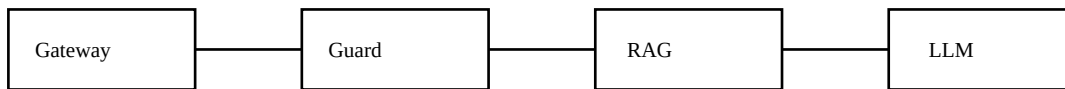
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AI Agent Instructions

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Page 19: Redis Architecture



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Redis Architecture**. Cursor AI should treat this chapter as authoritative when generating code.

- Rate Limiting
- Semantic Cache
- Distributed Locks
- Session Store

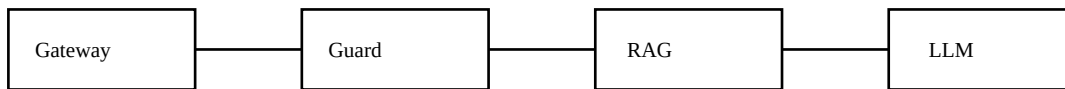
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Page 20: PostgreSQL Design



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **PostgreSQL Design**. Cursor AI should treat this chapter as authoritative when generating code.

- Users
- Conversations
- Evaluations
- Documents
- Audit Logs

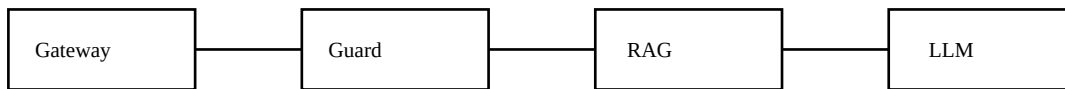
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AI Agent Instructions

Generate interfaces before implementations, keep configuration externalized, include README documentation and preserve service boundaries.

Page 21: Kafka Event Bus



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Kafka Event Bus**. Cursor AI should treat this chapter as authoritative when generating code.

- ai.requests
- ai.responses
- ai.security
- ai.evaluations
- Analytics Consumers

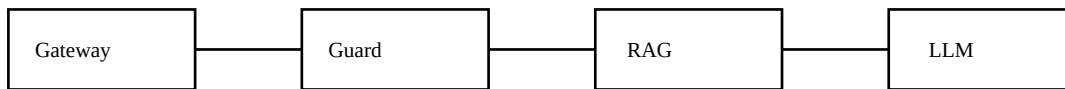
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AI Agent Instructions

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Page 22: Observability



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Observability**. Cursor AI should treat this chapter as authoritative when generating code.

- OpenTelemetry Traces
- Prometheus Metrics
- Grafana Dashboards
- Correlation IDs

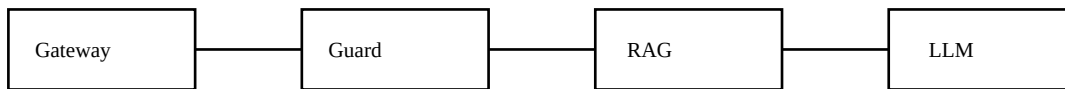
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AI Agent Instructions

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Page 23: API Specification



This chapter defines the architectural decisions, responsibilities and implementation boundaries for ****API Specification****. Cursor AI should treat this chapter as authoritative when generating code.

- POST /v1/chat
- POST /v1/evaluate
- GET /v1/health
- GET /v1/metrics

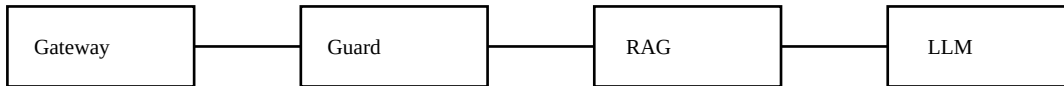
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*****AI Agent Instructions*****

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Page 24: CI/CD



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **CI/CD**. Cursor AI should treat this chapter as authoritative when generating code.

- GitHub Actions
- Container Builds
- Testing
- Quality Gates

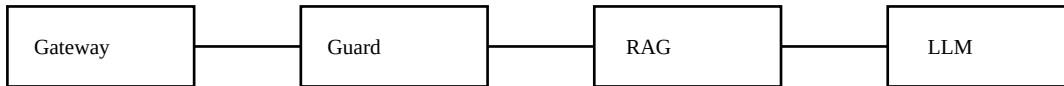
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AI Agent Instructions

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Page 25: Kubernetes Roadmap



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Kubernetes Roadmap**. Cursor AI should treat this chapter as authoritative when generating code.

- Ingress
- Autoscaling
- Rolling Updates
- Secrets
- ConfigMaps

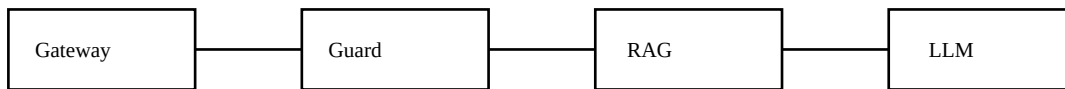
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Page 26: Enterprise Expansion



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Enterprise Expansion**. Cursor AI should treat this chapter as authoritative when generating code.

- Multi-tenancy
- Model Routing
- MCP Integration
- Human Approval
- Cost Optimization
- Governance

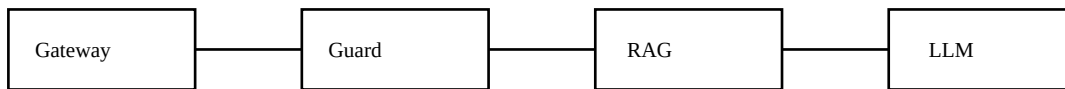
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AI Agent Instructions

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Page 27: Architecture Decision Records



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Architecture Decision Records**. Cursor AI should treat this chapter as authoritative when generating code.

- FastAPI over Spring Boot
- Redis over Memcached
- Kafka over RabbitMQ
- PostgreSQL over MongoDB

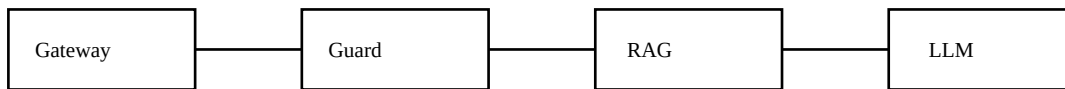
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Page 28: Threat Model



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Threat Model**. Cursor AI should treat this chapter as authoritative when generating code.

- Spoofing
- Tampering
- Repudiation
- Information Disclosure
- DoS
- Privilege Escalation

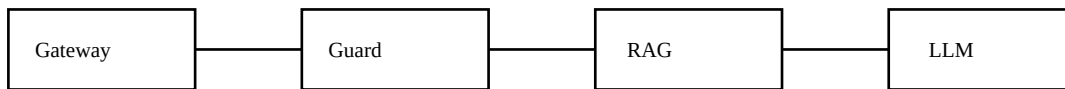
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AI Agent Instructions

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Page 29: Implementation Milestones



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Implementation Milestones**. Cursor AI should treat this chapter as authoritative when generating code.

- V1 Gateway
- V2 Observability
- V3 Enterprise Platform

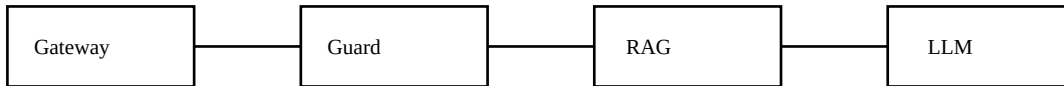
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Page 30: Interview Preparation



This chapter defines the architectural decisions, responsibilities and implementation boundaries for **Interview Preparation**. Cursor AI should treat this chapter as authoritative when generating code.

- Scaling Questions
- Failure Scenarios
- Tradeoffs
- Security Discussions

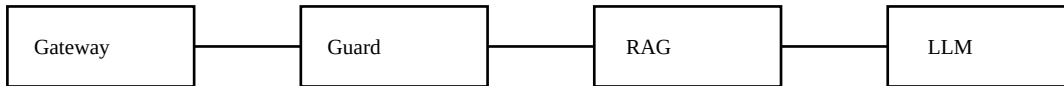
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Page 31: Cursor AI Kickstart Guide



This chapter defines the architectural decisions, responsibilities and implementation boundaries for ****Cursor AI Kickstart Guide****. Cursor AI should treat this chapter as authoritative when generating code.

- Generate gateway first.
- Generate auth service.
- Generate guardrail service.
- Generate RAG service.
- Generate worker.
- Generate infrastructure.
- Generate tests.

*****Acceptance Criteria*****

Every implementation generated from this chapter must remain Docker-compatible, expose health checks, structured logging, environment-based configuration and production-safe defaults.

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